

Practical Problems Ideas for scientific investigations often arise from practical problems involving humans and health or environmental issues. Salt marshes, for example, are vital nurseries for commercially important fish and shellfish. Yet salt marshes are under intense pressure from agriculture and from development. An engineer might wonder, "How would nearby construction affect the marsh and how can that impact be minimized?"

New Technology New technology often opens new ways of asking questions for both scientists and engineers. For example, portable, remote data-collecting equipment enables field researchers to monitor environmental conditions around the clock, in several different locations at once. This information allows researchers to pose and test new hypotheses.

 **READING CHECK Describe** Describe a "fact" that you have heard or read that made you skeptical. Explain your reasons for doubt.

Community Analysis and Feedback

Scientists often collaborate in groups and communicate with other research groups. In order for research to be accepted, however, it must be officially shared with the scientific community. The communication must follow a variety of rules that ensure it is scientifically appropriate. Some of the steps that scientists use when communicating their results are shown in **Figure 1-6**.

Scientists also share their work with the general public, especially when the results or information could benefit society. This communication is less formal, and can take the form of a newspaper or magazine article, television program, or blog post.



VIDEO

Listen to scientists discussing their work in extreme environments.

Figure 1-6

Community Analysis and Feedback

Communication is an important part of science. Scientists review and evaluate one another's work to ensure accuracy. Results from one study may lead to new ideas and further studies.



Adapted from *Understanding Science*, UC Berkeley, Museum of Paleontology